

MAUSAM IQ - TECHNICAL & API SPECIFICATION

Personalized Homepage for 'Mausam' Mobile & Web Application

Ministry of Earth Sciences (MoES) / India Meteorological Department (IMD) - SIH Architecture Blueprint

1. EXECUTIVE SUMMARY & CORE PARADIGM SHIFT

Traditional weather applications present static, scientific numbers (e.g. 'Precipitation 82%, Wind 28 km/h'). For ordinary citizens—farmers, urban commuters, delivery partners, and vulnerable citizens—raw numbers fail to answer: 'What does this weather mean for my life, and what action should I take right now?'

Mausam IQ transforms the IMD platform from an observational data feed into a Personalized Weather Intelligence Platform. It executes a deterministic 7-stage intelligence loop: RAW DATA -> WEATHER -> PERSONAL CONTEXT -> IMPACT -> SCENARIO -> ACTION -> TRANSPARENCY.

2. COMPLETE TECHNOLOGY STACK

The system is engineered as an enterprise-grade Universal Cross-Platform application running on Android, iOS, and Web:

| Layer                | Technology                | Version  | Architectural Role & Capability                                      |
|----------------------|---------------------------|----------|----------------------------------------------------------------------|
| Mobile & Web Core    | Universal React Native    | v0.86.3  | Native runtime bridge for mobile devices & adaptive web interface    |
| Cross-Platform SDK   | Expo SDK                  | v57.0.25 | Universal device abstraction layer, permissions & asset pipeline     |
| UI Rendering         | React 19 & React-DOM      | v19.2.3  | Concurrent rendering engine & component state lifecycle              |
| Web Compilation      | react-native-web          | v0.21.2  | Translates React Native primitives into optimized responsive Web DOM |
| Programming Language | TypeScript                | ~6.0.3   | Strict static compile-time type safety & domain interface contracts  |
| Styling & Tokens     | StyleSheet & Custom Theme | Native   | Zero-runtime overhead; Dark & Light mode; IMD 4-tier alert tokens    |
| Visual Effects       | expo-linear-gradient      | ~57.0.2  | Atmospheric backdrops, glassmorphism & radar visualization           |
| Iconography          | @expo/vector-icons        | v15.0.2  | MaterialCommunityIcons, Ionicons, and Feather symbols                |
| State Management     | React Context + Hooks     | Native   | Reactive state for Persona, Language, Location & Card Rankings       |
| Offline Storage      | AsyncStorage              | v2.2.0   | Local persistence of cache, user DNA & disaster warnings             |
| Internationalization | Custom i18n Engine        | v1.0.0   | Instant dynamic switching between English & Hindi (Bilingual)        |
| Bundling & Tooling   | Metro Bundler             | ~57.0.16 | High-speed hot module replacement & tree-shaking compiler            |

3. API ARCHITECTURE (11 ACTIVE APIS & SERVICES)

Mausam IQ operates with a total of 11 active APIs and service engines: 6 External Meteorological Feeds and 5 Internal Algorithmic Micro-Services.

A. External Meteorological & GIS Data Ingestion APIs (6 Feeds):

1. IMD Doppler Weather Radar (DWR) Telemetry API  
Function: Provides real-time ground radar reflectivity (dBZ), convective storm velocity, and localized cloud top heights from Palam, Safdarjung, and coastal radar network.
2. MoES / IMD Numerical Weather Prediction (NWP) Ensemble API  
Function: Delivers 24-hour hourly progression and 7-day high-resolution synoptic forecasts (Precipitation, Relative Humidity, Pressure, Wind Shear).
3. IMD Agromet Advisory Service (AAS) / ICAR API  
Function: Supplies hyperlocal agricultural advisories, crop vulnerability thresholds, evapotranspiration rates, and agrochemical spraying feasibility guidelines.
4. CPCB (Central Pollution Control Board) National AQI API  
Function: Streams real-time Air Quality Index metrics (PM2.5, PM10, NO2) with 6-tier health hazard classifications.
5. OpenStreetMap / CartoDB / WMO Vector Tile API  
Function: Renders interactive GIS radar maps, global meteorological wind streams (Jet Streams), and Inter-Tropical Convergence Zone (ITCZ) satellite overlays.
6. Open-Meteo & WMO Global Met Office Sync API  
Function: Synchronizes international weather feeds for global metropolises (Tokyo, London, New York, Dubai, Sydney, Cairo) for the global radar scope.

B. Internal Algorithmic Engine APIs (5 Core Micro-Services in src/services/):

1. weatherService (weatherService.ts)  
Function: fetchCurrentWeather(), fetchHourlyForecast(), fetch7DayForecast(), fetchActiveAlerts(), getSavedLocations(). Provides ground-truth data with confidence scores and freshness timestamps.

2. **impactEngine (impactEngine.ts)**

Function: analyzeWeatherImpact(). Converts raw weather facts into personalized citizen impact, calculates the 0-100 Weather Impact Score, factors breakdown, and interactive Action Checklists.

3. **routeEngine (routeEngine.ts)**

Function: getWhatIfScenarios(), calculateRouteWeather(). Simulates realistic departure windows ('Abhi Niklu' vs '7 PM' vs '9:30 PM' vs 'Kal Subah') with road waterlogging severity, delay minutes, and AI verdicts.

4. **personalizationEngine (personalizationEngine.ts)**

Function: rankHomepageCards(), getCardExplanation(). Multi-factor mathematical ranking algorithm scoring cards by Persona (0.35), Interests (0.25), Location (0.15), Time of Day (0.15), and Severe Alert Urgency (0.10).

5. **aiAssistant (aiAssistant.ts)**

Function: askMausamAI(). Grounded, zero-hallucination conversational meteorology copilot answering queries strictly based on IMD Doppler observations and official safety protocols.

4. CORE SIGNATURE USER-FACING MODULES

1. Weather -> Impact -> Action (Beyond Static Weather Data)

- Stage 1 (IMD Ground Reality): Displays official Doppler facts (e.g. Convective rain 32 mm/hr, squall 42 km/h).
- Stage 2 (Real-Life Citizen Impact): Automatically tailored to the active citizen persona (Commuter, Farmer, Delivery, Senior Citizen).
- Stage 3 (Action Playbook & Interactive Checklist): Provides concrete safety steps with a live tap-to-complete citizen action checklist (e.g. 'Clear field drainage channels', 'Switch to elevated metro lines').

2. Context-Specific Weather Impact Score & Factor Decomposition

- 0-100 Circular Dial Gauge: Dynamically color-coded (Low, Moderate, High, Severe) calibrated to user vulnerability.
- Contributing Vulnerability Factors: Transparent breakdown with exact percentage weights (e.g. Rain Intensity 35%, Underpass Waterlogging 30%, Wind Squalls 20%, Lightning 15%).
- Tap-to-Inspect Drawers: Tapping any factor reveals observed ground values and direct mitigation guidance.
- 'Why is my score High?' Modal: Mathematical calibration explanation.

3. What-If Simulator ('Abhi Niklu vs 7 PM vs Kal Subah' Comparison)

- Realistic Departure Windows: Compares 'Abhi Niklu (6:15 PM)' [Recommended, +8m delay], 'Sham 7:00 PM' [Avoid, +45m delay, severe underpass flooding], 'Raat 9:30 PM' [Safe delayed window], and 'Kal Subah (07:30 AM)' [Optimal, 0m delay].
- Dual Modes: Focus Deep-Dive View & Side-by-Side Matrix Table comparing Rain %, Delay, Waterlogging, and AI Verdicts.

4. Dual-Scope Radar Map (India National Radar + All Over The World Map)

- India Radar: Doppler composite, lightning alerts, cloud animation, and IMD coastal mesonet.
- World Map: Global metropolises (Tokyo, London, NYC, Dubai), Jet Streams, ITCZ rain belt, and timezones.

5. SIH EVALUATION & SYSTEM VERIFICATION

- Dev Server Endpoint: http://localhost:8081 (Universal Web / Responsive Desktop & Tablet).
- Metro Bundler: 200 OK Clean compilation (360+ modules).
- Zero Hallucination Guarantee: All AI responses are deterministically anchored to IMD ground telemetry.
- Offline Resilience: Critical disaster alerts and user DNA persist without active network connectivity.